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The Globalization Risk Premium

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1 Big picture

- **Question.** How is globalization reflected in asset prices? Do investors require compensation for holding firms exposed to foreign trade shocks?
- **Measurement.** The paper uses industry-level shipping costs as an inverse measure of globalization exposure. Low shipping costs mean goods can be imported cheaply, so domestic firms are more exposed to import competition and foreign productivity shocks.
- **Main asset-pricing result.** Firms in low-shipping-cost industries earn a large risk premium relative to firms in high-shipping-cost industries. The baseline long-low-SC, short-high-SC portfolio earns about 7% annually.
- **Mechanism.** The premium appears to come from displacement risk: when foreign firms become more productive, least efficient domestic firms in exposed industries suffer.
- **Macro-finance meaning.** Foreign productivity shocks must be bad states for domestic marginal investors if exposed firms command a premium.
- **Takeaway.** Globalization is not just an average productivity or consumer-price force. It also changes the covariance of firm cash flows with marginal utility and therefore affects expected returns.

2 Incremental contribution of the paper

1. **Trade exposure in asset prices.** The paper brings international trade into empirical asset pricing by asking whether exposure to globalization is priced.
2. **Shipping costs as exposure.** Rather than using realized imports alone, the paper uses persistent industry shipping costs and weight-to-value ratios to measure structural exposure to global competition.
3. **Risk-premium fact.** It documents a sizeable premium for stocks more exposed to globalization, robust to standard factor models and firm-characteristic controls.
4. **Displacement-risk channel.** The paper links the premium to import competition and the risk that foreign productivity shocks displace low-efficiency domestic firms.
5. **Asset-price discipline on trade models.** The calibration asks what risk-sharing and consumption responses are needed for a standard trade model to rationalize the observed premium.
6. **New interpretation of globalization.** The paper reframes globalization as a priced macroeconomic risk, not only as a source of gains from trade.

Interpretation: The incremental contribution is the combination of a new exposure measure, a new empirical risk premium, and a model-based interpretation of why foreign productivity shocks may be bad news for domestic investors.

3 Data and measurement

3.1 Shipping costs and weight-to-value

Shipping costs are measured at the industry-year level as

$$SC_{j,t} = \frac{CIF_{j,t} - FOB_{j,t}}{FOB_{j,t}},$$

where CIF is cost-insurance-freight import value and FOB is free-on-board value. The alternative proxy is a log weight-to-value ratio, measured as the weight in kilograms divided by import value.

Interpretation: Lower shipping costs and lower weight-to-value ratios imply lower trade frictions and greater exposure to foreign competition.

3.2 Trade, industry, and stock-market data

The industry sample covers 439 manufacturing industries. Trade and shipping data are merged with industry-level employment, shipments, value added, TFP, tariffs, imports, and exports. Stocks are assigned to industries, then sorted into portfolios by previous-year shipping costs or weight-to-value ratios.

Interpretation: The portfolios are built using predetermined exposure variables. The key high-exposure group is the *low* shipping-cost portfolio.

3.3 Main outcomes

The empirical tests focus on portfolio excess returns, factor-model alphas, earnings announcement returns, analyst forecast errors, and sensitivities to China import growth and import penetration. The model section uses simulated returns, consumption, profits, and trade quantities.

4 Empirical and model specifications

4.1 Trade-flow validation

The first step validates shipping costs as exposure to international trade:

$$TradeFlow_{j,t} = \alpha + \beta SC_{j,t} + \gamma Tariff_{j,t} + \delta' X_{j,t} + \lambda_t + \varepsilon_{j,t}.$$

A negative β for imports and exports means that high shipping costs reduce trade.

4.2 Portfolio returns

The main portfolio test estimates

$$R_{p,t}^e = \alpha_p + \beta_{MKT} MKT_t + \beta_{SMB} SMB_t + \beta_{HML} HML_t + \beta_{RMW} RMW_t + \beta_{CMA} CMA_t + \varepsilon_{p,t}.$$

A positive alpha for the low-SC minus high-SC portfolio indicates a globalization risk premium.

4.3 Earnings and analyst tests

To distinguish risk premia from predictable cash-flow surprises, the paper studies earnings announcement windows and analyst forecast errors. If high returns merely compensated for negative earnings surprises, low-SC portfolios should have systematically negative earnings-window returns or forecast errors. They do not.

4.4 Foreign-shock exposure

The paper links portfolios to import shocks using regressions of returns on China import growth and import penetration. Low-SC firms are more negatively exposed to these foreign shocks.

4.5 Model calibration

The final section uses a two-country trade model with asset pricing. The model simulates foreign productivity shocks, trade flows, domestic profits, domestic consumption, and stock returns under different assumptions about international risk sharing.

5 Identification and empirical design

- **Exposure design.** Shipping costs are persistent physical trade frictions that proxy for exposure to foreign competition.
- **Validation.** The paper first shows that low shipping costs predict more trade and more sensitivity to trade shocks.
- **Risk-premium design.** Portfolio sorts compare firms with different ex ante exposure to globalization.
- **Alternative explanations.** Standard equity factors, currency factors, firm characteristics, earnings surprises, and analyst forecast errors do not explain the premium.
- **Mechanism.** Low-SC firms load negatively on foreign import shocks, consistent with displacement risk.
- **Model discipline.** The calibrated model clarifies why the empirical premium requires foreign productivity shocks to be bad states for domestic investors.

Interpretation: The identification is a cumulative argument rather than a single natural experiment. The key empirical chain is: low shipping costs predict trade exposure; low-SC firms earn high returns; the premium is not explained by standard controls; low-SC firms are hurt by foreign shocks.

6 Tables and figures

6.1 Table I: Summary Statistics; Table II: Shipping Cost Persistence

Read: Table I reports industry-year variation in shipping costs, tariffs, imports, exports, employment, value added, shipments, and TFP. Table II shows persistence across shipping-cost and weight-to-value quintiles.

Economic message: The exposure variable is dispersed and persistent. This makes it meaningful for portfolio formation.

Table I
Summary Statistics

This table presents summary statistics for the industry-year sample that covers 439 unique manufacturing industries (with four-digit SIC codes between 2000 and 3999). Shipping costs are measured at the industry-year level as the percent difference between the cost-insurance-freight value and the free-on-board value of imports. Weight-to-value is measured at the industry-year level as the ratio of the weight in kilograms to the free-on-board value of imports. Tariffs are measured at the industry-year level as the ratio of customs duties to the free-on-board value of imports. Imports, exports, and net imports are measured at the industry-year level and normalized by the sum of total shipments and imports. Shipping costs, weight-to-value ratio, tariffs, imports, and exports are available from the Census and obtained from Peter Schott's website. Employment, shipments, value added, and total factor productivity (TFP) are obtained from the NBER-CES files and are available until 2011. All variables are winsorized at the first 99th percentiles. The sample period is 1974 to 2011.

	Obs.	Mean	Std. Dev.	p1	p50	p99
International Trade Data						
Shipping costs	14,366	0.056	0.038	0.002	0.047	0.224
Log weight-to-value	8,705	-1.750	1.519	-6.106	-1.742	2.154
Tariffs	14,366	0.043	0.051	0.000	0.027	0.261
Imports	14,366	0.169	0.192	0.000	0.099	0.887
Exports	14,366	0.106	0.117	0.000	0.067	0.619
Net Imports	14,366	0.062	0.196	-0.416	0.013	0.803

(a) Table I: summary statistics.

Shipping Cost Persistence

This table presents transition frequencies across shipping cost quintiles (weight-to-value quintiles) from year $t - 1$ to t in columns (1) to (6) and from year $t - 5$ to t in columns (7) to (12) in the sample over the period 1974 to 2014 (1989 to 2014). Shipping costs are measured at the industry-year level as the percent difference between the cost-insurance-freight value and the free-on-board value of imports. Weight-to-value is measured at the industry-year level as the ratio of the weight in kilograms to the free-on-board value of imports.

Panel A: Transition across Shipping Cost Quintiles											
	from year $t - 1$ to year t					from year $t - 5$ to year t					
	Q_1^t	Q_2^t	Q_3^t	Q_4^t	Q_5^t	Q_1^t	Q_2^t	Q_3^t	Q_4^t	Q_5^t	
Q_{t-1}^1	0.866	0.113	0.013	0.003	0.0061	Q_{t-5}^1	0.761	0.163	0.044	0.018	0.014
Q_{t-1}^2	0.113	0.734	0.137	0.014	0.002	Q_{t-5}^2	0.153	0.572	0.213	0.047	0.015
Q_{t-1}^3	0.010	0.138	0.685	0.158	0.010	Q_{t-5}^3	0.039	0.206	0.494	0.224	0.036
Q_{t-1}^4	0.004	0.012	0.154	0.709	0.121	Q_{t-5}^4	0.013	0.048	0.206	0.549	0.183
Q_{t-1}^5	0.003	0.005	0.014	0.120	0.857	Q_{t-5}^5	0.015	0.018	0.054	0.183	0.730

Panel B: Transition across Weight-to-Value Quintiles											
	from year $t - 1$ to year t					from year $t - 5$ to year t					
	Q_1^t	Q_2^t	Q_3^t	Q_4^t	Q_5^t	Q_1^t	Q_2^t	Q_3^t	Q_4^t	Q_5^t	
Q_{t-1}^1	0.946	0.047	0.003	0.001	0.002	Q_{t-5}^1	0.896	0.095	0.004	0.002	0.003
Q_{t-1}^2	0.047	0.882	0.065	0.005	0.000	Q_{t-5}^2	0.092	0.785	0.112	0.011	0.000
Q_{t-1}^3	0.002	0.068	0.863	0.066	0.001	Q_{t-5}^3	0.005	0.112	0.749	0.132	0.002
Q_{t-1}^4	0.002	0.003	0.066	0.884	0.046	Q_{t-5}^4	0.003	0.010	0.127	0.773	0.087
Q_{t-1}^5	0.002	0.001	0.002	0.045	0.951	Q_{t-5}^5	0.002	0.002	0.004	0.085	0.908

(b) Table II: shipping-cost and weight-to-value persistence.

6.2 Table III: Shipping Costs and Trade Flows

Read: Shipping costs and weight-to-value ratios are negatively related to imports and exports. The relation remains after controls and fixed effects.

Economic message: The table validates shipping costs as trade-exposure measures.

Table III Shipping Costs and Trade Flows									
resents results of industry-year regressions of the value of trade flows on shipping costs (Panel A) and the weight-to-value ratio (Panel B). Shipping costs are measured at the industry-year level as the percent difference between the cost-insurance-freight value and the free-on-board value of imports. Weight-to-value is measured at the industry-year level as the ratio of customs duties to the free-on-board value of imports. Some regressions are for the industry level of tariffs, penetration, log employment, log value added, log shipments, and total factor productivity (TFP), from the NBER-CES data sets. Standard errors are clustered at the industry level and reported in parentheses. *, **, and *** indicate the 10%, 5%, and 1% level, respectively. The sample period is 1974 to 2011 in Panel A, and 1989 to 2011 in Panel B.									
Panel A: Shipping Costs									
Imports			Exports			Net Imports			
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	
s	-0.327*	-0.614***	-0.096	-0.690***	-0.707***	-0.030	0.369*	-0.099	-0.059
	(0.168)	(0.160)	(0.091)	(0.119)	(0.106)	(0.096)	(0.169)	(0.119)	
		0.643***	-0.556***		-0.242***	-0.126	0.877***	-0.431***	
		(0.137)	(0.141)		(0.055)	(0.078)	(0.139)	(0.147)	
ent		0.031***	-0.067***		-0.031***	-0.028*	0.061***	-0.040*	
		(0.012)	(0.016)		(0.010)	(0.015)	(0.013)	(0.021)	
led		-0.042*	-0.054***		0.023	-0.009	-0.064***	-0.045*	
		(0.023)	(0.020)		(0.015)	(0.018)	(0.022)	(0.026)	
s		-0.038*	-0.009		-0.003	0.008	-0.034	-0.013	
		(0.022)	(0.022)		(0.013)	(0.018)	(0.022)	(0.027)	
		0.021	-0.006		0.011	-0.008	0.017	0.009	
		(0.037)	(0.020)		(0.025)	(0.018)	(0.034)	(0.024)	
	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	
	No	No	Yes	No	No	No	Yes	No	
	14,366	14,366	14,366	14,366	14,366	14,366	14,366	14,366	
	0.143	0.327	0.861	0.129	0.163	0.741	0.049	0.262	0.783

(Continued)

(a) Table III, Panel A: shipping-cost regressions.

Table III—Continued									
Panel B: Weight-to-Value Ratio									
Imports			Exports			Net Imports			
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	
-0.040***	-0.037***	-0.003	-0.025***	-0.032***	0.006	-0.015**	-0.005	-0.009	
(0.006)	(0.006)	(0.007)	(0.004)	(0.004)	(0.005)	(0.007)	(0.006)	(0.008)	
	1.284***	-0.196		-0.755***	-0.256**		2.016***	0.073	
	(0.273)	(0.157)		(0.104)	(0.102)		(0.274)	(0.201)	
	0.014	-0.056***		-0.043***	-0.007		0.056***	-0.049**	
	(0.016)	(0.016)		(0.010)	(0.015)		(0.016)	(0.024)	
	0.056	-0.038		0.007	-0.005		-0.044	-0.011	
	(0.031)	(0.018)		(0.016)	(0.019)		(0.030)	(0.029)	
	-0.037	-0.051**		0.017	-0.014		-0.053*	-0.033	
	(0.033)	(0.021)		(0.015)	(0.023)		(0.031)	(0.035)	
	0.056	-0.028		0.009	-0.025		0.044	-0.008	
	(0.041)	(0.020)		(0.027)	(0.020)		(0.039)	(0.029)	
	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	
	No	No	Yes	No	Yes	No	No	Yes	
	8,705	8,705	8,705	8,705	8,705	8,705	8,705	8,705	
	0.132	0.383	0.934	0.105	0.187	0.828	0.031	0.333	0.871

(b) Table III, Panel B: weight-to-value regressions.

6.3 Table IV: Portfolio Summary Statistics; Table V: Portfolio Returns

Read: Table IV shows that low-SC portfolios earn higher average excess returns than high-SC portfolios. Table V shows that this return spread survives factor adjustments.

Economic message: These are the core empirical risk-premium results.

Table IV Shipping Cost and Weight-to-Value Portfolios: Summary Statistics						
This table reports summary statistics for five shipping cost portfolios (Panel A) and five weight-to-value portfolios (Panel B) based on U.S. stocks traded on the Amex, NASDAQ, or NYSE. Shipping costs are measured at the industry-year level as the percent difference between the cost-insurance-freight value and the free-on-board value of imports. Weight-to-value is measured at the industry-year level as the ratio of the weight in kilograms to the free-on-board value of imports. ME is the average portfolio market capitalization over the sample period converted into 2013 constant billions dollars. BE/ME is book-to-market equity, defined as book value of equity (item CEQ) divided by market value of equity (item CSHO × item PRCC_F). ROA is defined as operating income after depreciation and amortization (item OIBDP-itemDP) divided by total assets. I/K is capital expenditures (item CAPX) divided by property, plants, and equipment (item PPENT). Market leverage is total debt (item DLC + item DLTT) divided by the sum of total debt and market value of equity. The sample period is 1975 to 2015 in Panel A, and 1990 to 2015 in Panel B.						
Panel A: Shipping Cost Portfolios						
	Low	2	3	4	High	Hi-Lo
<i>Portfolio Characteristics</i>						
ME	4.494	3.185	2.889	3.781	4.184	
BE/ME	0.589	0.659	0.697	0.805	0.939	
Market leverage	0.150	0.155	0.177	0.228	0.299	
ROA	-0.064	0.026	0.040	0.081	0.096	
I/K	0.311	0.320	0.295	0.247	0.200	
<i>Portfolio Moments</i>						
Mean excess return (%)	16.053	9.763	9.150	10.081	9.065	-6.988
Sharpe ratio	0.562	0.386	0.390	0.465	0.466	-0.350
Panel B: Weight-to-Value Portfolios						
	Low	2	3	4	High	Hi-Lo
<i>Portfolio Characteristics</i>						
ME	5.341	4.035	2.777	3.346	6.084	
BE/ME	0.427	0.491	0.538	0.604	0.732	
Market leverage	0.096	0.093	0.113	0.194	0.279	
ROA	-0.151	-0.028	0.006	0.063	0.088	
I/K	0.336	0.337	0.321	0.255	0.179	

(a) Table IV: shipping-cost and weight-to-value portfolio characteristics.

Table V Shipping Cost and Weight-to-Value Portfolios: Returns													
This table presents excess returns (a) over a five-factor Fama-French model of either shipping cost portfolios (Panel A) or weight-to-value portfolios (Panel B). Monthly returns are multiplied by 12 to make the magnitude comparable to annualized returns. Shipping costs are measured at the industry-year level as the percent difference between the cost-insurance-freight value and the free-on-board value of imports. In any given month, the returns are sorted into five portfolios based on either their industry shipping costs or weight-to-value ratio in the previous year. We regress a given portfolio's return in excess of the risk-free rate on the market portfolio minus the risk-free rate, the size factor (small minus big), the value factor (high minus low), the profitability factor (profit minus weak), and the investment factor (conservative minus aggressive), all obtained from Kenneth French's website. Portfolio returns are either equally weighted (columns 1 to 6) or value-weighted (columns 7 to 12). Standard errors are in parentheses and estimated using Newey-West with 12 lags. ***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively. The sample is 1975 to 2015 in Panel A and 1990 to 2015 in Panel B.													
Panel A: Shipping Cost Portfolios													
Equally Weighted						Value-Weighted							
Low	2	3	4	High	Hi-Lo	Low	2	3	4	High	Hi-Lo		
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)		
9.702***	1.245	1.074	-1.103	-4.005***	-13.707***	4.244***	0.592	0.658	-0.067	-0.954	-5.198**		
(2.359)	(1.695)	(1.646)	(1.524)	(1.379)	(3.019)	(1.536)	(1.610)	(1.695)	(1.546)	(1.247)	(2.073)		
1.064***	1.043***	1.018***	1.128***	1.091***	0.027	0.990***	1.022***	1.053***	1.089***	0.936***	-0.054		
(0.033)	(0.026)	(0.026)	(0.032)	(0.034)	(0.066)	(0.046)	(0.038)	(0.026)	(0.026)	(0.026)	(0.048)		
-0.620***	-0.305***	-0.204***	0.263***	0.503***	1.123***	-0.536***	-0.268***	-0.068	-0.071	0.161*	0.697***		
(0.074)	(0.061)	(0.070)	(0.092)	(0.096)	(0.125)	(0.052)	(0.075)	(0.084)	(0.063)	(0.091)	(0.109)		
0.950***	1.039***	0.884***	0.789***	0.730***	-0.220***	-0.111**	0.159***	0.052	0.104	0.013	0.124*		
(0.061)	(0.057)	(0.063)	(0.061)	(0.053)	(0.079)	(0.049)	(0.064)	(0.072)	(0.065)	(0.049)	(0.072)		
-0.821***	-0.445***	-0.479***	-0.149	0.211**	1.033***	-0.013	-0.139	-0.468***	-0.060	0.306***	0.319***		
(0.109)	(0.068)	(0.084)	(0.104)	(0.089)	(0.150)	(0.072)	(0.091)	(0.113)	(0.081)	(0.064)	(0.094)		
0.153	0.010	0.015	-0.080	-0.040	-0.195	0.145	-0.036	-0.167	0.146	0.289***	0.153		
(0.159)	(0.108)	(0.128)	(0.167)	(0.115)	(0.216)	(0.162)	(0.170)	(0.116)	(0.102)	(0.111)	(0.198)		

(Continued)

(b) Table V, Panel A: U.S. factor-adjusted portfolio returns.

Table V—Continued

Panel B: Weight-to-Value Portfolios

Equally Weighted					Value-Weighted				
Low (1)	2 (2)	3 (3)	4 (4)	High (5)	Low (6)	2 (7)	3 (8)	4 (9)	High (10)
11.320*** (3.559)	4.743* (2.793)	2.756 (2.385)	-1.862 (1.993)	-3.238* (1.713)	-14.558*** (4.238)	4.414** (1.967)	5.670*** (2.074)	1.277 (1.795)	-2.633 (1.637)
1.060*** (0.046)	1.034*** (0.037)	1.063*** (0.039)	1.092*** (0.039)	1.121*** (0.048)	0.062 (0.077)	0.902*** (0.060)	1.074*** (0.050)	1.127*** (0.046)	1.060*** (0.031)
-0.626*** (0.098)	-0.350*** (0.062)	-0.208** (0.092)	0.349*** (0.100)	0.642*** (0.084)	1.268*** (0.118)	-0.447*** (0.069)	-0.483*** (0.089)	-0.412*** (0.062)	0.223*** (0.086)
0.948*** (0.098)	1.014*** (0.090)	0.868*** (0.087)	0.731*** (0.074)	0.657*** (0.051)	-0.291*** (0.110)	-0.130 (0.090)	0.142 (0.089)	0.154** (0.064)	0.171*** (0.042)
-0.898*** (0.162)	-0.552*** (0.084)	-0.524*** (0.105)	-0.096 (0.092)	0.127** (0.062)	1.025*** (0.199)	-0.014 (0.116)	-0.317*** (0.074)	-0.295* (0.155)	0.208*** (0.061)
0.248 (0.187)	-0.064 (0.127)	-0.027 (0.180)	-0.104 (0.179)	-0.078 (0.122)	-0.526 (0.262)	0.235 (0.173)	-0.324*** (0.103)	-0.007 (0.152)	0.111 (0.079)
									0.130 (0.130)
									0.262 (0.262)

(a) Table V, continuation: additional portfolio-return estimates.

Table VI

Shipping Cost and Weight-to-Value Portfolios: Evidence from European Stock Markets

Table VI presents excess returns (a) over a five-factor Fama-French model of either shipping costs portfolios (Panel A) or weight-to-value portfolios (b). Monthly returns are multiplied by 12 to make the magnitude comparable to annualized returns. In any given month, stocks traded on stock markets are sorted into five shipping cost and five weight-to-value portfolios based on the same sorting of four-digit SIC code industries (Table V). Monthly returns and four-digit SIC codes are both obtained from the EUROINDAT database. As in Fama and French (2012), monthly returns are in U.S. dollars and monthly excess returns are returns in excess of the one-month U.S. Treasury bill rate. We include all stocks (2,601 stocks in manufacturing industries for which data on shipping costs are available) traded in the following 16 European countries: Austria, Germany, Switzerland, Denmark, Spain, Finland, France, Great Britain, Greece, Ireland, Italy, the Netherlands, Norway, Portugal, and Sweden. This list of European countries is the one studied in Fama and French (2012) and used to compute the five factors for Europe (the market is minus the risk-free rate, the size factor, the value factor, the profitability factor, and the investment factor), all available on Kenneth French's website from July 1990. Portfolio returns are either equally weighted (columns (1) to (6)) or value-weighted (columns (7) to (12)). Standard errors are in parentheses and estimated using Newey-West with 12 lags. ***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively. The sample period is 1990 to 2015.

Panel A: Shipping Cost Portfolios

Equally Weighted					Value-Weighted				
Low (1)	2 (2)	3 (3)	4 (4)	High (5)	Low (6)	2 (7)	3 (8)	4 (9)	High (10)
7.888*** (2.806)	1.809 (2.863)	2.436 (2.048)	-0.218 (1.467)	-2.034* (1.094)	-9.923*** (3.227)	8.461*** (2.513)	3.524 (3.989)	0.784 (3.401)	-2.549 (2.813)
0.963*** (0.043)	1.158*** (0.040)	1.026*** (0.035)	1.084*** (0.025)	1.010*** (0.017)	0.027 (0.048)	0.799*** (0.054)	1.108*** (0.076)	1.082*** (0.058)	1.039*** (0.036)
-0.463*** (0.115)	0.052 (0.137)	-0.176 (0.118)	0.184** (0.091)	0.340*** (0.056)	0.803*** (0.138)	-0.600*** (0.154)	0.063 (0.241)	0.098 (0.296)	0.095 (0.081)
0.613*** (0.072)	0.711*** (0.068)	0.705*** (0.055)	0.780*** (0.045)	0.743*** (0.037)	0.139* (0.071)	-0.338*** (0.148)	0.052 (0.145)	0.116 (0.137)	0.169** (0.077)
-0.671*** (0.128)	-0.350*** (0.169)	-0.434*** (0.089)	-0.094 (0.050)	0.054 (0.140)	0.725*** (0.195)	-0.232 (0.257)	-0.448* (0.225)	-0.170 (0.181)	0.494*** (0.118)
-0.261 (0.155)	-0.237* (0.137)	-0.211** (0.105)	-0.144* (0.083)	-0.098 (0.068)	0.153 (0.180)	-0.004 (0.247)	-0.478* (0.249)	-0.509*** (0.148)	0.095 (0.098)
									0.214 (0.180)
									0.218 (0.271)

(b) Table VI: evidence from European markets, first part.

Table VI—Continued

Panel B: Weight-to-Value Portfolios

Equally Weighted					Value-Weighted				
Low (1)	2 (2)	3 (3)	4 (4)	High (5)	Low (6)	2 (7)	3 (8)	4 (9)	High (10)
3.354*** (2.946)	4.133 (2.598)	1.450 (2.073)	1.050 (1.422)	-2.379** (1.113)	-10.733*** (3.438)	5.569* (3.039)	15.015*** (3.757)	2.943 (2.465)	-0.332 (2.578)
1.005*** (0.047)	1.118*** (0.037)	1.036*** (0.040)	1.066*** (0.022)	1.011*** (0.015)	0.006 (0.051)	0.749*** (0.050)	1.076*** (0.062)	1.110*** (0.063)	1.105*** (0.058)
3.991*** (1.111)	-0.221** (0.111)	-0.219* (0.126)	0.176*** (0.064)	0.380*** (0.063)	0.771*** (0.139)	-0.278*** (0.093)	-0.729*** (0.211)	-0.426*** (0.144)	0.541*** (0.123)
3.613*** (0.071)	0.620*** (0.076)	0.754*** (0.057)	0.768*** (0.048)	0.747*** (0.033)	0.134* (0.072)	-0.197** (0.080)	-0.386** (0.173)	0.146 (0.097)	0.092 (0.107)
0.642*** (0.142)	-0.430*** (0.148)	-0.525*** (0.176)	-0.063 (0.058)	0.060 (0.055)	0.702*** (0.141)	-0.117 (0.171)	-0.728*** (0.323)	0.379** (0.197)	0.428*** (0.125)
0.128 (0.153)	-0.419*** (0.105)	-0.265* (0.147)	-0.088 (0.059)	-0.113 (0.072)	0.015 (0.188)	0.465*** (0.117)	-0.743*** (0.295)	-0.205 (0.191)	-0.499*** (0.132)
									0.256 (0.163)
									0.218 (0.158)

Table VI, continuation: European portfolio returns.

6.4 Table VII: Earnings Announcement Returns; Table VIII: Analyst Forecast Errors

Read: Earnings-window returns and analyst forecast errors do not reveal systematic bad cash-flow surprises for low-SC firms.

Economic message: The premium is unlikely to be explained by predictable earnings disappointment.

Table VII
Shipping Cost and Weight-to-Value Portfolios: Returns around Earnings Announcements

Table presents the returns around earnings announcements of stocks sorted into either shipping cost portfolios (Panel A) or weight-to-value (Panel B). The table reports both cumulative excess returns (stock return minus the risk-free rate) over a seven-day window (11-day w) from five days (10 days) prior to the quarterly earnings announcement day to one day after the announcement day, which we refer to as the window (−10, 1) window). These announcement returns are then either equally weighted (columns (1) to (6)) or value-weighted (column (7)) for each calendar quarter at the level of the shipping cost and weight-to-value portfolios. When value-weighted, we weight each return by its market capitalization measured at the end of the calendar quarter prior to the earnings announcement. Shipping costs are measured at industry-year level as the percent difference between the cost-insurance-freight value and the free-on-board value of imports. Weight-to-value is measured at the industry-year level as the ratio of the weight in kilograms to the free-on-board value of imports. Standard errors (in parentheses) are based on the time series of quarterly returns, and estimated using Newey-West with 12 lags. ***, **, and * indicate significance at the 1%, 5%, 10% level, respectively. The sample period is 1975 to 2015 in Panel A, and 1990 to 2015 in Panel B.

Panel A: Shipping Costs Portfolios												
Equally Weighted						Value-Weighted						
Low	2	3	4	High	Hi-Lo	Low	2	3	4	High	Hi-Lo	
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	
Increment Ret. (−5, 1)	0.364	0.337**	0.443***	0.484***	0.431***	0.167	0.357*	0.334	0.496**	0.370*	0.502***	0.146
	(0.207)	(0.164)	(0.133)	(0.128)	(0.116)	(0.195)	(0.187)	(0.217)	(0.185)	(0.189)	(0.100)	(0.193)
Increment Ret. (−10, 1)	0.615**	0.528**	0.763***	0.747***	0.647***	0.032	0.530**	0.288	0.688***	0.604**	0.521***	−0.009
	(0.304)	(0.225)	(0.210)	(0.189)	(0.180)	(0.271)	(0.266)	(0.264)	(0.253)	(0.263)	(0.197)	(0.248)
Panel B: Weight-to-Value Portfolios												
Equally Weighted						Value-Weighted						
Low	2	3	4	High	Hi-Lo	Low	2	3	4	High	Hi-Lo	
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	
Increment Ret. (−5, 1)	0.406	0.564***	0.567***	0.590***	0.591***	0.185	0.563***	0.724***	1.163***	0.590***	0.401***	−0.162
	(0.279)	(0.192)	(0.181)	(0.135)	(0.158)	(0.298)	(0.145)	(0.183)	(0.182)	(0.189)	(0.103)	(0.148)
Increment Ret. (−10, 1)	0.818***	0.827***	0.935***	0.847***	0.870***	0.052	0.819***	0.790**	1.130***	0.798***	0.653***	−0.165
	(0.406)	(0.295)	(0.293)	(0.204)	(0.243)	(0.446)	(0.223)	(0.291)	(0.392)	(0.233)	(0.211)	(0.270)

(a) Table VII: returns around earnings announcements.

6.5 Table IX: Exporter Double Sorts; Table X: China Import Growth Betas

Read: Table IX examines whether the premium is explained by exporter status or firm-level export exposure. Table X links portfolio returns to China import growth and import penetration.

Economic message: The evidence points toward import-competition and displacement risk rather than a pure exporter-benefit channel.

Table IX
Shipping Cost and Weight-to-Value Portfolios: Returns, Conditional on Size and Profitability

Table presents excess returns (a) over a five-factor Fama-French model of either shipping cost portfolios (Panel A) or weight-to-value portfolios (Panel B). Monthly returns are multiplied by 12 to make the magnitude comparable to annualized returns. Shipping costs are measured at the y -year level as the percent difference between the cost-insurance-freight value and the free-on-board value of imports. Weight-to-value is measured at the industry-year level as the ratio of the weight in kilograms to the free-on-board value of imports. In any given month, stocks are identically sorted into five portfolios based on either their industry shipping costs or weight-to-value ratio in the previous year, and into three or based on either their market capitalization (Size) in the previous month or their return on assets (ROA) in year $t - 2$. Stocks at the intersection of the two sorts are grouped together to form portfolios based on shipping costs and either size or ROA in Panel A and based on weight-to-value and either size or ROA in Panel B. We then regress a given portfolio's return in excess of the risk-free rate on the market portfolio minus the a rate, the size factor (small minus big), the value factor (high minus low), the profitability factor (robust minus weak), and the investment conservative minus aggressive, all obtained from Kenneth French's website. Portfolio's returns are either equally weighted (columns (1) to value-weighted (columns (7) to (12)). Standard errors are in parentheses and estimated using Newey-West with 12 lags. ***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively. The sample period is 1975 to 2015 in Panel A and 1990 to 2015 in Panel B.

Panel A: Shipping Costs Portfolios												
Equally Weighted						Value-Weighted						
Low	2	3	4	High	Hi-Lo	Low	2	3	4	High	Hi-Lo	
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	
Size Tertiles						Size Tertiles						
12.033***	0.064	2.002	−1.079	−6.402***	−18.436***	10.204***	0.505	2.682	−0.749	−4.864***	−17.788***	
	(4.498)	(2.488)	(2.379)	(2.393)	(2.069)	(3.136)	(0.803)	(2.311)	(3.922)	(1.864)	(3.361)	
9.534***	2.105	−0.323	−1.584	−3.730**	−13.285***	9.555***	2.029	−0.103	−1.146	−4.067**	−13.621***	
	(0.605)	(2.311)	(1.830)	(1.518)	(1.077)	(3.421)	(2.973)	(2.090)	(1.619)	(1.490)	(1.602)	(3.460)
9.005***	1.389	1.800	−0.978	−3.139***	−12.333***	4.176***	0.772	0.801	0.072	−0.965	−5.081**	
	(2.129)	(1.454)	(1.865)	(1.473)	(1.223)	(3.102)	(1.357)	(1.682)	(1.746)	(1.598)	(1.393)	(2.986)
												(3.189)
ROA Tertiles						ROA Tertiles						
10.187***	−0.415	0.199	−0.694	−6.419***	−16.605***	10.579***	−0.159	4.734	1.909	−6.675***	−17.203***	
	(3.166)	(2.471)	(2.386)	(2.691)	(3.857)	(2.790)	(2.359)	(2.874)	(3.062)	(3.106)	(3.978)	
10.991***	3.328	1.996	−0.130	−0.573*	−13.584***	8.962***	−0.410	−4.067**	−1.861	−0.728	−8.938***	
	(2.265)	(1.512)	(1.873)	(1.447)	(1.480)	(2.048)	(2.038)	(2.089)	(1.991)	(1.507)	(2.782)	
9.977***	2.808*	2.421	−0.098	−1.643	−8.646***	2.708	0.019	1.095	0.869	−0.682	−3.448	
	(1.789)	(1.560)	(1.679)	(1.330)	(1.330)	(2.339)	(2.090)	(2.568)	(2.964)	(1.658)	(2.922)	(3.806**)
												(4.640)

(Continued)

(a) Table IX, part 1: double sorts related to exporter exposure.

Table VIII
Analysts' Forecast Errors

Table reports the coefficients from panel regressions of either the actual $I/B/E/S$ annual earnings per share (EPS) (columns (1), (4), (7), and (10)), or $I/B/E/S$ consensus forecast of annual EPS (columns (2), (5), (8), and (11)), or the forecast error (actual $I/B/E/S$ annual EPS minus mean consensus forecast of annual EPS; columns (3), (6), (9), and (12)), all normalized by the stock price at the end of the last fiscal year, on either $\ln$ of the weight-to-value ratio, and control variables. The consensus forecast is measured as the average of the last 10 of each analyst covering the stock in the eight months before the end of the fiscal year. Shipping costs are measured at the industry-year level as the percent difference between the cost-insurance-freight value and the free-on-board value of imports. Weight-to-value is measured at the industry-year level as the ratio of the weight in kilograms to the free-on-board value of imports. BETA for a stock in a given month is the beta of the stock returns with the U.S. stock market return estimated using monthly data over the past 60 months. LNME is the logarithm of firm capitalization in the previous month. BEAME is book-to-market equity, defined as book value of equity (item CEQ) divided by market value y (item CSHO + item PRCC_F) at the end of fiscal year $t - 2$. Return on assets (ROA) is defined as operating income after depreciation and amortization (item OIBDP + item DPT) divided by total assets at the end of fiscal year $t - 2$. I/K is capital expenditures (item CAPX) divided by property, plant, and equipment (item PPENT) at the end of fiscal year $t - 2$. MARKET LEV is total debt (item DLTCL + item DLTTL) divided by the sum of total debt and market value of equity at the end of fiscal year $t - 2$. We remove observations for which the forecast error is below the first and above the 99th percentiles of its empirical distribution. Standard errors (in parentheses) are clustered at the four-digit industry level. ***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively. The sample period is 1982 to 2015.

Equally Weighted						Value-Weighted						
Actual	Forecast	Error	Actual	Forecast	Error	Actual	Forecast	Error	Actual	Forecast	Error	
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	
0.671***	0.658***	0.013	0.329*	0.323	0.006	0.016***	0.016***	0.000	0.011**	0.011**	0.000	
	(0.242)	(0.245)	(0.014)	(0.198)	(0.196)	(0.005)	(0.005)	(0.000)	(0.005)	(0.005)	(0.000)	
Weight-to-value						−0.018***	−0.016***	−0.001**	−0.016***	−0.015***	−0.001	
						(0.003)	(0.003)	(0.001)	(0.004)	(0.004)	(0.001)	
						0.020***	0.018***	0.002***	0.021***	0.018***	0.002***	
						(0.003)	(0.003)	(0.000)	(0.003)	(0.003)	(0.000)	
						0.008	0.009	−0.001	0.012	0.013	−0.001	
						(0.010)	(0.010)	(0.001)	(0.009)	(0.009)	(0.001)	
ST LEV						0.017	0.028*	−0.008***	−0.008	−0.001	−0.007***	
						(0.014)	(0.013)	(0.002)	(0.015)	(0.015)	(0.002)	
						−0.048***	−0.045***	−0.003**	−0.035***	−0.033***	−0.002**	
						(0.009)	(0.009)	(0.001)	(0.011)	(0.011)	(0.001)	
ROA	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	
ROA	22.915	22.915	22.915	22.915	22.915	22.915	22.915	22.915	22.915	22.915	22.915	
ROA	0.031	0.037	0.021	0.081	0.082	0.044	0.045	0.050	0.016	0.089	0.037	

(b) Table VIII: analyst forecast errors.

Table IX—Continued

Table presents excess returns (a) over a five-factor Fama-French model of either shipping cost portfolios (Panel A) or weight-to-value portfolios (Panel B). Monthly returns are multiplied by 12 to make the magnitude comparable to annualized returns. Shipping costs are measured at the y -year level as the percent difference between the cost-insurance-freight value and the free-on-board value of imports. Weight-to-value is measured at the industry-year level as the ratio of the weight in kilograms to the free-on-board value of imports. In any given month, stocks are identically sorted into five portfolios based on either their industry shipping costs or weight-to-value ratio in the previous year, and into three or based on either their market capitalization (Size) in the previous month or their return on assets (ROA) in year $t - 2$. Stocks at the intersection of the two sorts are grouped together to form portfolios based on shipping costs and either size or ROA in Panel A and based on weight-to-value and either size or ROA in Panel B. We then regress a given portfolio's return in excess of the risk-free rate on the market portfolio minus the a rate, the size factor (small minus big), the value factor (high minus low), the profitability factor (robust minus weak), and the investment conservative minus aggressive, all obtained from Kenneth French's website. Portfolio's returns are either equally weighted (columns (1) to value-weighted (columns (7) to (12)). Standard errors are in parentheses and estimated using Newey-West with 12 lags. ***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively. The sample period is 1975 to 2015 in Panel A and 1990 to 2015 in Panel B.

Panel B: Weight-to-Value Portfolios												
Equally Weighted						Value-Weighted						
Low	2	3	4	High	Hi-Lo	Low	2	3	4	High	Hi-Lo	
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	
Size Tertiles						Size Tertiles						
16.469***	5.575	4.736	−0.398	−5.445**	−21.913***	15.101***	6.579*	4.399	−0.536	−5.754**	−20.850***	
	(5.708)	(3.718)	(3.884)	(3.125)	(2.783)	(5.324)	(3.764)	(3.686)	(3.205)	(2.531)	(5.847)	
9.789**	4.943	1.065	−2.745	−3.875**	−13.456***	10.364***	4.368	1.770	−3.568*	−4.273**	−14.658***	
	(3.944)	(3.391)	(2.622)	(1.843)	(1.821)	(4.501)	(3.861)	(3.297)	(2.401)	(1.831)	(1.800)	(4.452)
9.645***	4.609**	2.503	−2.920	−2.289	−11.802***	4.479**	6.120***	1.372	−2.524	0.320	−4.158	
	(2.677)	(2.108)	(1.917)	(1.780)	(1.650)	(3.445)	(2.040)	(2.144)	(1.872)	(1.660)	(1.490)	(2.909)
												(6.698**)
												(5.854)
ROA Tertiles						ROA Tertiles						
15.430***	4.329	2.512	−2.109	−6.097**	−18.527***	8.722***	2.963	6.645*	−0.736	−5.235	−19.897***	
	(4.656)	(4.058)	(3.795)	(3.312)	(2.953)	(5.372)	(3.314)	(3.890)	(3.692)	(3.790)	(3.190)	(4.660)
10.900***	7.050***	3.196	−0.791	−2.437	−15.811***	11.132***	0.644	1.518	−6.911***	0.766	−10.368**	
	(2.941)	(2.106)	(2.015)	(1.908)	(1.892)	(3.960)	(3.207)	(2.902)	(2.472)	(2.902)	(1.964)	(4.354)
6.522**	3.271**	4.864*	−1.698	−1.723	−8.949***	2.440	7.746***	1.871	−1.406	−1.127	−3.567	
	(5.816)	(2.301)	(3.039)	(1.728)	(1.434)	(3.140)	(2.380)	(2.688)	(2.194)	(1.857)	(1.932)	(3.562)
												(10.390*)
												(5.060)

(b) Table IX, part 2: continuation.

Chinese Import Growth Betas: Conditional on Size and Profitability

Table X presents Chinese import growth betas of each shipping cost portfolios (columns (1) to (6)) and weight-to-value portfolios (columns (7) to (12)). Shipping costs are measured at the industry-year level as the percent difference between the cost-insurance-freight value and the free-on-board value of imports. Weight-to-value is measured at the industry-year level as the ratio of the weight in kilograms to the free-on-board value of imports. In any given month, stocks are independently sorted into five portfolios based on either their industry shipping costs or weight-to-value ratio in the previous month or based on their return on assets in year $t - 2$. Stocks at the intersection of the two sorts are grouped together to form portfolios based on shipping costs and either size or profitability. We then compute Chinese import growth betas for each (double-sorted) portfolio as the coefficient β of the following OLS regression estimated at the monthly frequency over the sample period 1974 to 2015: $R_{i,t}^{PW} = \beta_J \cdot \text{ChImpGr}_t + \alpha_J + u_{J,t}$, where $R_{i,t}^{PW}$ is the equally weighted portfolio excess return in month t and ChImpGr_t is the growth rate excess imports to the United States between month t and the same month in the previous year. Standard errors are in parentheses and are reported using Newey-West with 12 lags. ***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively. The sample period is 1974 to 2015 in columns (1) to (6), and 1989 to 2015 in columns (7) to (12).

Chinese (Univariate) Import Growth Betas											
Shipping Cost Portfolios						Weight-to-Value Portfolios					
Low (1)	2 (2)	3 (3)	4 (4)	High (5)	Hi-Lo (6)	Low (7)	2 (8)	3 (9)	4 (10)	High (11)	Hi-Lo (12)
-0.629* (0.334)	-0.404 (0.297)	-0.329 (0.278)	-0.262 (0.253)	-0.162 (0.213)	0.406\$* (0.231)	-0.737* (0.436)	-0.324 (0.400)	-0.432 (0.375)	-0.238 (0.297)	-0.191 (0.271)	0.487 (0.319)
Size Terciles						Size Terciles					
-0.742* (0.391)	-0.642* (0.330)	-0.504* (0.302)	-0.353 (0.274)	-0.255 (0.248)	0.407 (0.272)	-0.961* (0.524)	-0.573 (0.430)	-0.543 (0.402)	-0.317 (0.335)	-0.281 (0.314)	0.591 (0.393)
-0.607* (0.361)	-0.373 (0.325)	-0.270 (0.300)	-0.237 (0.272)	-0.345 (0.240)	0.256 (0.261)	-0.697 (0.462)	-0.226 (0.436)	-0.513 (0.406)	-0.230 (0.327)	-0.262 (0.307)	0.401 (0.354)
-0.425 (0.286)	-0.183 (0.280)	-0.205 (0.268)	-0.206 (0.243)	-0.057 (0.204)	0.356 (0.218)	-0.387 (0.374)	-0.165 (0.370)	-0.242 (0.372)	-0.177 (0.277)	-0.094 (0.252)	0.251 (0.291)
ROA Terciles						ROA Terciles					
-0.717* (0.390)	-0.569 (0.354)	-0.500 (0.330)	-0.186 (0.328)	-0.140 (0.278)	0.548\$* (0.285)	-0.893* (0.501)	-0.483 (0.484)	-0.677 (0.442)	-0.186 (0.406)	-0.016 (0.351)	0.789\$** (0.382)
-0.497* (0.300)	-0.223 (0.266)	-0.201 (0.245)	-0.253 (0.248)	-0.191 (0.215)	0.283 (0.220)	-0.469 (0.395)	-0.020 (0.342)	-0.305 (0.347)	-0.251 (0.306)	-0.185 (0.274)	0.156 (0.316)
-0.484* (0.293)	-0.345 (0.294)	-0.378 (0.241)	-0.293 (0.226)	-0.151 (0.197)	0.354\$* (0.181)	-0.554* (0.329)	-0.297 (0.344)	-0.404 (0.327)	-0.249 (0.261)	-0.189 (0.240)	0.346 (0.230)

Table X: China import growth and foreign-shock exposure.

6.6 Tables XI-XII: Additional Return Tests

Read: The additional tables examine portfolio returns under alternative sorts and factor models. They confirm that the premium is robust to richer controls.

Economic message: The premium is not a disguised exposure to standard equity factors.

Table XI Shipping Cost Portfolios: Returns, Conditional on U.S. Trade Elasticities (σ)											
Panel A: Small Size Only											
Equally Weighted						Value-Weighted					
Low (1)	2 (2)	3 (3)	4 (4)	High (5)	Hi-Lo (6)	Low (7)	2 (8)	3 (9)	4 (10)	High (11)	Hi-Lo (12)
1.843 (5.017)	2.730 (3.320)	3.056 (2.199)	-1.770 (2.280)	-4.407** (1.809)	-6.450* (2.176)	1.586 (2.830)	2.732 (2.360)	2.798 (2.130)	-1.789 (2.219)	-5.746** (1.810)	-5.333 (3.400)
14.105*** (3.940)	1.494 (2.690)	1.935 (2.454)	1.047 (2.699)	-5.403* (2.820)	-19.508*** (4.753)	12.602*** (3.555)	1.222 (2.991)	3.015 (2.555)	1.498 (2.449)	-5.227* (2.671)	-17.869*** (4.884)
Low σ					-13.397*** (4.966)						-12.566** (5.090)
Panel B: Low ROA Only											
Equally Weighted						Value-Weighted					
Low (1)	2 (2)	3 (3)	4 (4)	High (5)	Hi-Lo (6)	Low (7)	2 (8)	3 (9)	4 (10)	High (11)	Hi-Lo (12)
1.664 (2.796)	-0.012 (3.471)	3.075 (2.509)	-2.705 (1.856)	-4.719** (1.856)	-6.383* (3.353)	4.689 (3.450)	0.023 (3.485)	4.553** (3.026)	-4.750 (3.026)	-0.707 (1.817)	-5.396 (3.860)
12.029*** (3.532)	2.848 (2.287)	0.262 (2.195)	3.075 (2.508)	-5.724** (2.838)	-17.752*** (4.840)	8.650*** (2.114)	1.622 (2.249)	4.408 (2.844)	4.420 (2.844)	-2.263** (3.472)	-16.118*** (4.420)
Low σ					-11.269** (4.883)						-10.722* (5.548)

(a) Table XI: characteristic-sorted portfolio tests.

Table XII Shipping Cost Portfolios: Returns, Conditional on Pareto Parameter (γ)											
Panel A: Small Size Only											
Equally Weighted						Value-Weighted					
Low (1)	2 (2)	3 (3)	4 (4)	High (5)	Hi-Lo (6)	Low (7)	2 (8)	3 (9)	4 (10)	High (11)	Hi-Lo (12)
8.864*** (2.973)	-1.218 (2.579)	2.316 (2.976)	-2.911 (2.381)	-4.227* (2.215)	-13.061*** (3.840)	8.451*** (2.760)	0.037 (2.802)	2.582 (3.967)	-1.805 (2.186)	-2.389 (2.053)	-10.850*** (4.690)
16.328*** (4.365)	2.266 (2.677)	1.364 (2.354)	-0.313 (2.386)	-1.217*** (1.992)	-21.542*** (4.764)	15.709*** (4.205)	2.380 (2.288)	1.485 (2.288)	-0.993 (2.352)	-4.994*** (3.812)	-20.703*** (4.848)
Low γ					-6.432** (3.575)						-9.853** (5.051)
Panel B: Low ROA Only											
Equally Weighted						Value-Weighted					
Low (1)	2 (2)	3 (3)	4 (4)	High (5)	Hi-Lo (6)	Low (7)	2 (8)	3 (9)	4 (10)	High (11)	Hi-Lo (12)
9.930*** (2.712)	-1.906 (1.951)	0.989 (2.840)	-2.512 (2.820)	-4.673** (2.033)	-13.703*** (3.605)	7.902*** (2.455)	-0.280 (2.275)	-1.006 (2.374)	-3.144 (3.307)	-1.804 (1.969)	-9.837*** (2.294)
12.929*** (3.924)	3.988 (2.575)	0.888 (2.301)	-0.006 (2.200)	-6.213*** (2.303)	-18.142*** (4.796)	14.612*** (4.291)	1.931 (2.491)	-0.295 (2.548)	-4.128* (2.460)	-8.864*** (2.634)	-23.476*** (5.011)
Low γ					-4.439 (4.023)						-13.635** (6.397)

(b) Table XII: Fama-French five-factor alphas.

6.7 Tables XIII-XIV: Calibration and Model Moments

Read: Table XIII reports calibrated model parameters. Table XIV reports simulated moments for high- and low-exposure industries.

Economic message: The model shows which risk-sharing conditions are needed to rationalize the empirical premium.

Table XIII
Calibrated Parameters

This table presents values of the parameters used in our model. Dynamic parameters are calibrated at a quarterly frequency. We also indicate whether the parameters are obtained from the literature, from our own estimates, or from calibration that match specific moments of the data. Aggregate quantities for China and the United States are from the World Bank (data.worldbank.org/indicator).

Parameter	Symbol	Value	Source
Preferences (dynamic):			
Subjective Discount rate	β	0.99	Bansal and Yaron (2004)
Elasticity of intertemporal substitution	ψ	1.5	Bansal and Yaron (2004)
Relative risk aversion	ν	20	
Industry Organization:			
Manufacturing expenditure shares	a_0, a_0^*	0.1–0.9	
Elasticity of consumer demand	σ_J	3.8	Ghironi and Melitz (2005)
Elasticity across industries	θ	1.2	Loualiche (2015)
Pareto tail parameter	γ_J	3.4	Ghironi and Melitz (2005)
Production Technology:			
Labor supply	L, L^*	1–3	Ratio of working age population (China to U.S.)
Mass of firms in each industry	M_J	1	Average import penetration
	M_J^*	30–15	
Trade:			
Iceberg costs	τ_J	1–1.5	Ghironi and Melitz (2005)
Exporting fixed costs	f_J, f_J^*	$5\text{--}3 \cdot 10^{-5}$	Fraction of exporters
Aggregate Fluctuations:			
Domestic productivity process	μ_A	7	Ratio of GDP per capita (U.S. to China)
	σ_A	1.6%	
	ρ_A	0.97	
Foreign productivity process	μ_{A^*}	1	China Import to the U.S.
	σ_{A^*}	6%	
	ρ_{A^*}	0.98	

Under no risk-sharing ($\Xi = 0$), the wealth effect is negative and dominates. In that case, domestic consumption responds negatively to (positive) foreign

(a) Table XIII: calibrated parameters.

Table XIV
Model Simulation: Key Moment Conditions

This table reports key moments from simulations of our model. The model is simulated for one million periods under the shock processes described in the calibration Table XIII. Values in italics are derived from our own empirical estimates in the actual data. Standard errors are in parentheses and estimated using Newey-West with 12 lags. Ranges across industries are in brackets.

Quantities by Sectors				
	High Exposure Industry		Low Exposure Industry	
Import Penetration (I_J)				
Average	27%	(27%)	6%	(4.9%)
Std. deviation	15.4%	(12.6%)	5.3%	(2.9%)
cov($A^*, \cdot$)	0.49		0.16	
cov($A, \cdot$)	−0.09		−0.03	
Domestic Profits ($\pi_{D,J}$)				
cov($A^*, \cdot$)	−0.88		−0.39	
cov($A, \cdot$)	0.17		0.08	
Fraction of Exporters (ζ_J)				
Average	4%	[5%–30%]	3%	[5%–30%]
Aggregate Quantities				
	Aggregate Consumption (C_t)		Risk-Free Rate (Annualized)	
Average	-		3.7%	(4.7%)
Std. deviation	6%	(2.7%)	0.4%	(3.4%)
cov($A^*, \cdot$)	−0.2		1.1	
cov($A, \cdot$)	0.05		−0.34	
Average Excess Returns				
	High Exposure Industry		Low Exposure Industry	
Domestic Firms				
Average excess returns	1.04%	(16%)	0.54%	(9%)
Std. deviation	10.5%	(8.6%)	5.6%	(5.6%)
cov($\varepsilon_{A^*}, \cdot$)	−1.72		−0.92	
cov($\varepsilon_A, \cdot$)	0.26		0.16	
Average Exporters				
Average excess returns	0.49%		0.34%	
Std. deviation	4.9%		3.5%	
cov($\varepsilon_{A^*}, \cdot$)	0.70		0.56	
cov($\varepsilon_A, \cdot$)	−0.02		−0.02	

(b) Table XIV: model simulation moments.

6.8 Figures 1-2: Impulse Responses to Foreign Productivity Shocks

Read: Figure 1 compares domestic consumption, prices, and returns with and without risk sharing. Figure 2 focuses on import penetration, exporter shares, and profits without risk sharing.

Economic message: Foreign productivity shocks can lower exposed domestic profits and, depending on risk sharing, occur in bad domestic consumption states.

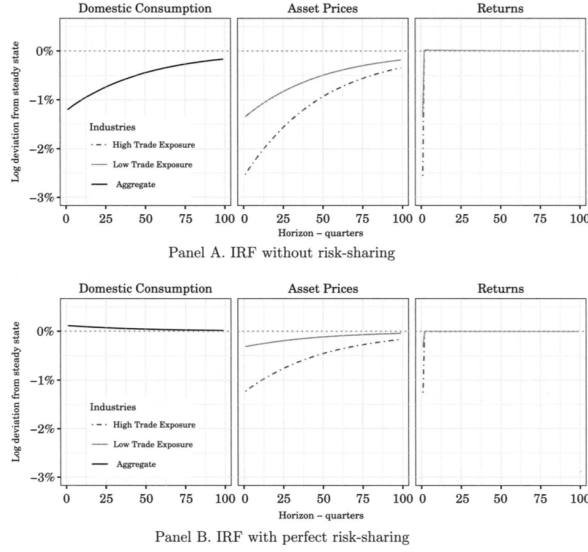


Figure 1. Impulse response: shock to A^* with (Panel A) and without risk-sharing (Panel B). This figure plots the impulse response function (IRF) to a shock ε^{A^*} from 2,000 model simulations. Quantities are log-deviations from their nonstochastic steady-state values. Domestic consumption and foreign consumption are C_t and C_t^* in the model, respectively. Import penetration is $I_{J,t}$, the fraction of exporters is $\zeta_{J,t}$, and domestic profits is $\pi_{D,J,t}$. Red lines correspond to industries with low trade costs that are more exposed to foreign competition. Blue lines are industries with higher trade costs. (Color figure can be viewed at wileyonlinelibrary.com)

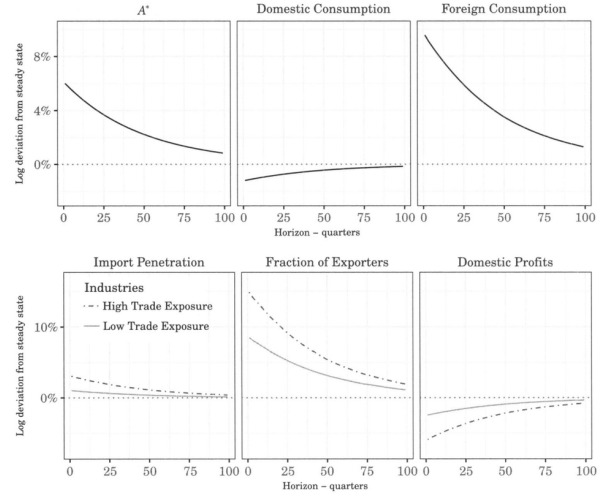


Figure 2. Impulse response: shock to A^* without risk-sharing. This figure plots the Impulse Response Function (IRF) to a shock ε^{A^*} from 2,000 model simulations. Quantities are log-deviations from their nonstochastic steady-state values. Domestic consumption and foreign consumption are C_t and C_t^* in the model, respectively. Import penetration is $I_{J,t}$, the fraction of exporters is $\zeta_{J,t}$, and domestic profits is $\pi_{D,J,t}$. Red lines correspond to industries with low trade costs that are more exposed to foreign competition. Blue lines are industries with higher trade costs. (Color figure can be viewed at wileyonlinelibrary.com)

(a) Figure 1: impulse responses to a foreign productivity shock. (b) Figure 2: additional impulse responses without risk sharing.

6.9 Figure 3: Sensitivity Analysis - Degree of Risk Sharing

Read: Figure 3 plots average returns and consumption responses as the degree of international risk sharing changes.

Economic message: The globalization premium is sensitive to risk sharing; full risk sharing can attenuate the domestic marginal-utility channel.

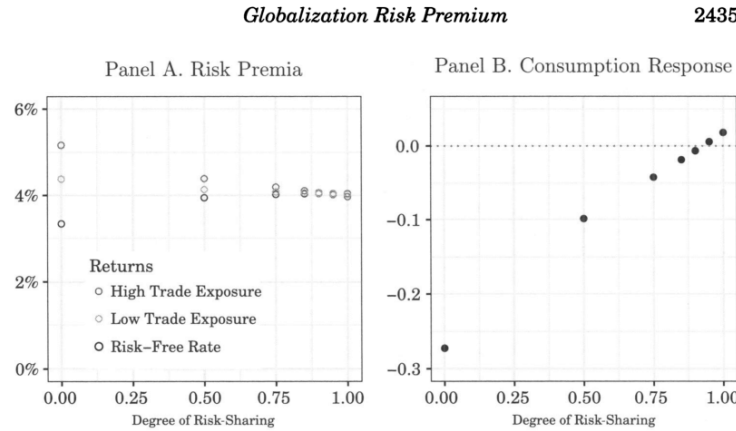


Figure 3. Sensitivity analysis: degree of risk-sharing. In this figure, we simulate the model for one million periods. In Panel A, we estimate average returns for different values of the risk-sharing parameter Ξ between zero and one. We include the risk-free rate to illustrate how returns exposed to trade act as a hedge with sufficient risk-sharing. In Panel B, we show the elasticity of consumption to shocks to foreign productivity A^* . (Color figure can be viewed at wileyonlinelibrary.com)

Figure 3: sensitivity analysis over the degree of risk sharing.

7 Short summary

- Shipping costs measure exposure to globalization.
- Low shipping costs imply high exposure to foreign productivity shocks and import competition.
- Low-SC firms earn higher average returns than high-SC firms.
- The premium survives standard factor and characteristic controls.
- Earnings announcements and analyst forecast errors do not explain the premium.
- Low-SC firms are more negatively exposed to import shocks, especially China import growth.
- The model shows that a premium arises when foreign productivity shocks are bad states for domestic investors.

8 Conclusion

8.1 Core conclusion

The paper shows that globalization exposure is priced in equity markets. Firms in industries with low shipping costs earn high average returns because they are exposed to foreign shocks that are bad for domestic marginal investors.

8.2 Economic intuition

Foreign productivity growth increases import competition. Exposed domestic firms, especially less efficient firms, suffer from displacement risk. If these shocks also coincide with low domestic consumption or high marginal utility, investors demand compensation to hold exposed firms.

8.3 Incremental contribution revisited

The paper's main contribution is to merge trade and asset pricing. It uses a physical measure of trade exposure to construct equity portfolios, documents a globalization risk premium, and then uses a trade model to ask what risk-sharing environment can rationalize that premium.

8.4 Limitations and open questions

Shipping costs are not randomized. The model produces a smaller premium than the empirical estimate, suggesting that additional mechanisms such as incomplete markets, labor-market frictions, disaster risk, or household heterogeneity may amplify the premium.

8.5 Takeaway

Globalization affects not only average real outcomes but also risk allocation. Asset prices reveal that exposure to foreign trade shocks is a priced source of risk.